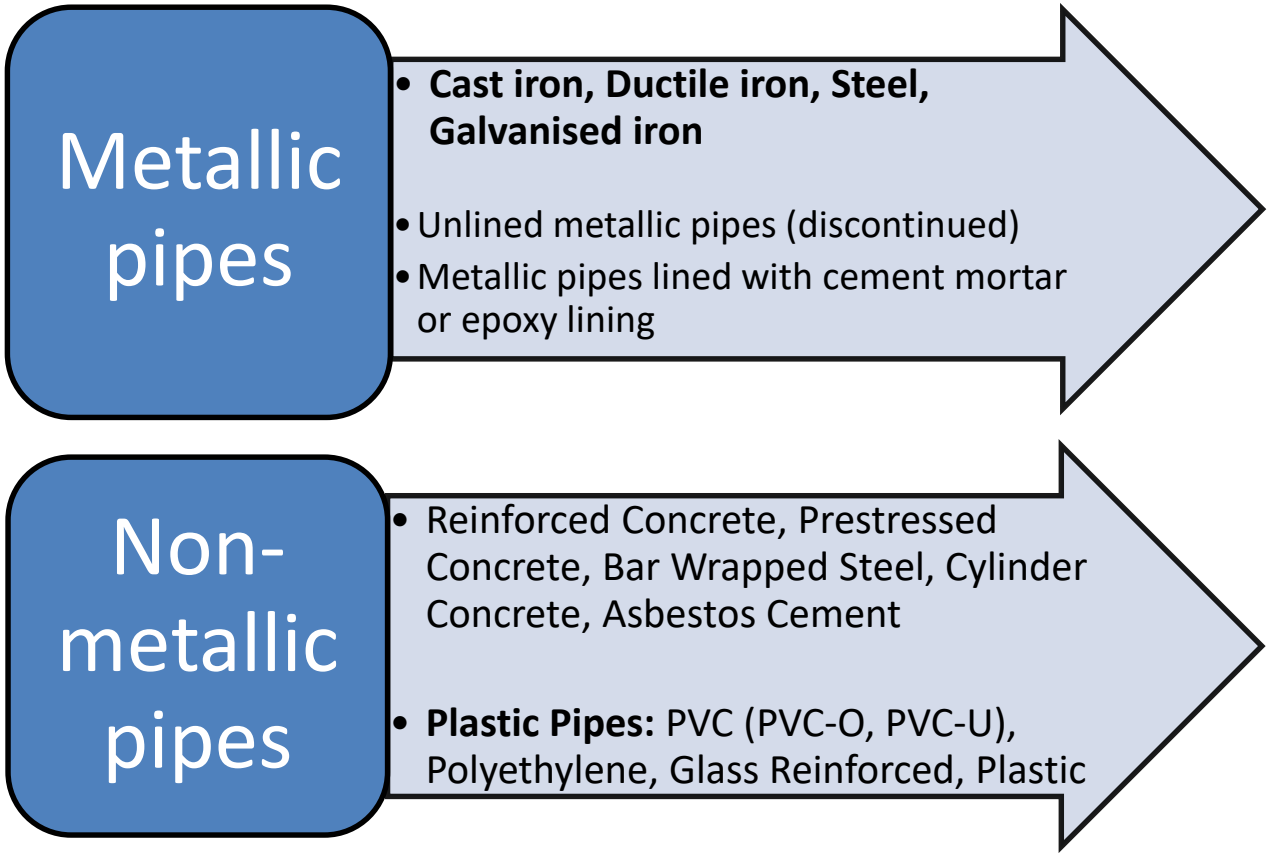
# Elements of Water Supply: Pipes, Joints, Valves

Pipes represent a large proportion of the capital invested in water supply undertakings and therefore are of particular importance. Therefore, pipe materials shall have to be judiciously selected not only from the point of view of durability, life and overall cost which includes, besides the pipe cost, the installation and maintenance costs necessary to ensure the required function and performance of the pipeline throughout its designed life time.



# Metallic Pipes

- Cast iron pipes
- Ductile iron pipes
- Galvanised pipes
- Steel pipes

# Cast Iron Pipes

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- Available in diameters from 80 mm to 1050 mm and are covered with protective coatings
- supplied in 3.66 m and 5.5 m lengths
- A variety of joints are available including socket, spigot, and flanged joints.
- Classified as LA, A and B according to their thickness. Class LA pipes have been taken as the basis for evolving the series of pipes.

- Application: They are well suited for pressure mains and laterals where tapings are made for house connections.

- Advantages: a) Good lasting qualities; b) Good strength, strong and durable; c) Good corrosion resistance if coated can be used in soils and for water of slightly aggressive character; d) Well suited for pressure mains and laterals where tapping is made for house connections

- Disadvantages: a) Heavy weight, and consequent high transport costs; b) Short length, leading to higher laying and jointing cost; c) Low tensile strength; d) Liability to defect of inner surface; e) Likely to break during transportation and jointing



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# Ductile Iron Pipes

- made by a metallurgical process, which involves the addition of magnesium into molten iron of low sulphur content.
- The ductile iron pipes are usually provided with cement mortar lining at the factory by centrifugal process to ensure a uniform thickness throughout its length
- available in the range of 80 mm to 1000 mm diameter, in lengths of 5.5 to 6 m
- Advantages: a) High resistance against breakage due to impact; b) High tensile strength, comparable to that of mild steel so that the pipes can be used; c) for higher working pressure; d) Traditional corrosion resistance, comparable to that of cast iron; e) Lighter in mass as compared to cast iron pipes
- Disadvantages: a) Heavy in weight and consequent difficulties in transportation & laying & jointing; b) Poor adaptability in laying & jointing; c) Poor response to surge pressures



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# Galvanised Iron Pipes

- manufactured using mild steel strips of Low Carbon
- available in tubes of 6 mm nominal bore to 150 mm nominal bore. The different classes of tubes like Light, Medium and Heavy depending on wall thickness, are distinguished by color bands such as Light Tubes (Yellow), Medium Tubes (Blue) and Heavy Tubes (Red)
- Application: generally used for distribution of treated or raw water in rural or hilly regions due to its light weight and easy jointing; It is not recommended to use GI pipes for house service connections as the pipe material is highly corrosive
- Advantages: a) Higher durability and longevity; b) Weld consistency and integrity; c) Amenable to rigorous fabrication; d) Superior finish and anti - rust coating; e) Greater corrosion resistance ; f) Superior bend ability and ease of cutting and threading
- Disadvantages: a) Galvanized iron should never be used underground unless properly covered which can be inconvenient for many jobs; b) It often hides significant defects beneath the zinc coating on the steel; c) Galvanized iron pipes may contain lead, which corrodes quickly and reduces the lifespan of the piping; d) Galvanized iron may leave rough patches inside pipes, resulting in serious failures and stoppages that can be expensive to repair



# Steel Pipes

- manufactured with the steel produced by the open hearth or electric furnace or one of the basic oxygen processes
- The pipes shall be manufactured by the processes of Seamless pipes, electric resistance welded pipes, submerged arc welded pipes
- Steel pipes of smaller diameter can be made from solid bar sections by hot or cold drawing processes and these tubes are referred to as seamless



- Advantages: a) Highly elastic; b) Strength; c) Easy installation; d) High flow capacity; e) Leak resistance; f) Long service life; Reliability; Versatility; Economical and cost-effective

- Disadvantages: a) Corrosion is the biggest limitation for any metallic pipe material; b) Welding joints for steel water pipe are complicated and require skilled labour; c) Steel pipe is susceptible to internal tuberculation and external corrosion, and is subjected to electrolysis, if not properly protected; d) Use of anti-corrosion products increases the price of production and maintenance of steel pipe; e) Air vacuum valves are necessary in large diameter pipes to prevent collapse, which adds to budgeted costs of installation

# Non-metallic Pipes

- Asbestos cement pipes
- Reinforced cement concrete pipes
- Prestressed concrete pipes
- Bar wrapped steel cylinder concrete pressure pipes

# Asbestos Cement Pipes

- made of a mixture of asbestos paste and cement compressed by steel rollers to form a laminated material of great strength and density
- manufactured from classes 5 to 25 and nominal diameters of 80 mm to 600 mm with the test pressure of 5 to 25 Kg/cm$^2$



- Advantages: a) The inside surface of pipe is smooth; The joining of pipes is very good and flexible; The pipes are anti-corrosive and cheap in cost; Light in weight to handle and transport
- Disadvantages: a) It is not resistant to impact damage; b) It cannot be located with standard line location equipment; c) It is difficult to locate using specialized location equipment; d) It does not transmit sound vibrations well, making the pinpointing of leaks very difficult; e) Has possible health concerns associated with the ingestion of asbestos; f) Exposure to airborne asbestos when lines are repaired or replaced

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# Reinforced Cement Concrete Pipes

- RCC pipes used in water supplies are classified as P1, P2 and P3 with test pressures of 2.0, 4.0, and 6.0 Kg/cm2 respectively.
- For use as gravity mains, the working pressure should not exceed 2/3 of the test pressure. For use as pumping mains, the working pressure should not exceed half of the test pressure.

- Advantages: a) Good corrosion resistance; b) Widespread availability; c) High strength; d) Good load supporting capacity

- Disadvantages: a) Require careful installation to avoid cracking; b) Heavy; c) Susceptible to attack from aggressive soils; d) Poor adaptability in installation



# Prestressed Concrete Pipes

- PSC pipes are suited for water supply mains where pressures in the range of 6 kg/cm2 to 20 kg/cm2 are encountered
- Two types of PSC pipes are in use:
- i) Cylinder type: Lined cylinder pipe commonly available in inside diameters ranging from 16 to 48 in.; Embedded cylinder pipe has been manufactured larger than 250 in. in diameter; Lengths are generally 16 - 24 ft
- ii) Non cylinder type: Consists of a concrete core which is pre-compressed both in longitudinal and circumferential directions by a highly stressed wire

- Advantages: a) Good corrosion resistance; b) Widespread availability; c) High strength; d) Good load supporting capacity

- Disadvantages: a) Require careful installation to avoid cracking; b) Heavy; c) Susceptible to attack from aggressive soils; d) Poor adaptability in installation



# Bar Wrapped Steel Cylinder Concrete Pressure Pipes

- available in diameters of 250 mm to 1600 mm and higher diameter pipes can be designed for working pressures upto 25 kgs/cm2
- Effective length of pipes shall be 4 m to 8 m. Longer length pipes can also be custom made



- Advantages: a) Pipes have semi-rigid or semi-metallic properties and resistant to impact; b) Stiffer than conventional steel pipes; c) Less expensive bedding cost than and side support required; d) Corrugated surface increases its structural stability; e) Cheaper and low maintenance pipes

- Disadvantages: a) Heavier than pipes like DI, HDPE, DWC, PVC, GRP pipes which makes it difficult to handle; b) Rough handled may damage the outer coat or inner lining

# Plastic Pipes

- PVC pipes (polyvinyl chloride)
- UPVC pipes (unplasticized polyvinyl chloride)
- OPVC pipes (oriented polyvinyl chloride)
- CPVC pipes (chlorinated polyvinyl chloride)
- Polyethylene pipes
- HDPE pipes (high density polyethylene)
- MDPE pipes (medium density polyethylene)
- GRP pipes (glass fibre reinforced plastic)

# PVC Pipes (polyvinyl chloride )

- Rigid PVC pipes weigh only 1/5th of conventional steel pipes of comparable sizes. PVC
- pipes are available in sizes of outer dia. 20, 25, 32, 50, 63, 75, 90, 110, 140, 160, 250, 290, and 315mm at working pressures of 2, 5 ,4, 6, 10 Kg/cm2

- Advantages: a) Resistance to corrosion; Light weight; b) Toughness; c) Rigidity; d) Economical in laying, jointing and maintenance; e) Ease of fabrication

- Disadvantages: a) Deteriorating effect by direct sunlight; b) Water hammer causes stretching of the pipe; c) Non-conductive material



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# UPVC Pipe (unplasticized polyvinyl chloride )

- The UPVC pipe is a plastic pipe made of polyvinyl chloride (PVC) resin and containing no plasticizer
- Pipes for supplying drinking water are mostly made of polyethylene (PE) or polyvinyl chloride (PVC). Unplasticised PVC pipes are greatly used for transportation of water
- Advantages: a) Lightweight; b) Easy to install; c) Economical; d) Good corrosion resistance; e) Smooth surface reduces friction losses; f) Long pipe sections reduce leakage/ infiltration potential; g) Flexible
- Disadvantages: a) Susceptible to chemical attack especially solvents; b) Strength affected by sunlight; c) Not suitable for above ground installations; d) Require great care during laying; e) Susceptible to damages due to external pressure and blows in above ground level application

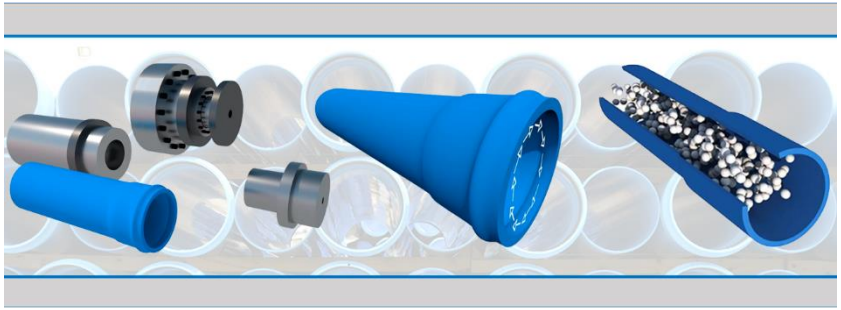


# OPVC Pipe (oriented polyvinyl chloride )

- Under certain conditions of pressure, temperature, and speed by stretching the material, it is possible to orient the PVC polymer molecules in the same direction. The result is a plastic with a layered structure called as Oriented Poly Vinyl Chloride Pipes of highest Orientation Class 500 with homogeneous Socket including supply of Elastomeric sealing ring
- manufactured for the different sizes in India ie., 110mm, 160mm, 200mm, 250mm, 315mm and 400mm with pressure ratings of PN 12.5, PN 16 and PN 25
- Application: conveyance of cold water under pressure and is suitable for conveyance of water, including potable water, up to and including 45$^{o}$C; in those applications where special performance requirements are needed, such as impact loads and pressure fluctuations, up to pressure of 2.5 MPa

Advantages: a) Greater lightness and easy handle; b) Excellent flexibility; c) Greater hydraulic capacity; d) Higher chemical resistance; e) Higher hydrostatic resistance; f) Higher resistance against water hammers

Disadvantages: a) Susceptible to chemical attack especially solvents; b) Strength affected by sunlight; c) Not suitable for above ground installations



# CPVC Pipe (chlorinated polyvinyl chloride )

- chlorinated polyvinyl chloride polymer from which the pipe compound is to be manufactured shall have chlorine content > 66.5 %
- chlorinated polyvinyl chloride pipe compounds containing additives such as modifiers, lubricants, fillers, etc., from which the pipes are to be manufactured, shall have a density between 1450 kg/m3 and 1650 kg/m3
- Nominal size of CPVC pipe is varying from 15 mm to 150 mm
- The wall of the plain pipe shall not transmit more than 0.1 percent of the visible light falling on it when tested in accordance with {IS 12235 (Part 3): 2004, Reaffirmed 2019}

Advantages: a) Lightweight and easy for transportation; b) Requires fewer tools for installation and maintenance; c) Lack of plasticizers which discourages microbial growth; d) Corrosion and abrasion resistance; e) Reground into pellets and recycled; f) Reduces heat loss due to lower thermal conductivity

Disadvantages: a) High thermal expansion coefficient; b) Slightly costlier than other PVC pipes



# HDPE (High Density Polyethylene) Pipes

- HDPE pipe is a type of flexible plastic pipe used for water supply systems and is made from the high density polyethylene) which is suitable for high pressure pipelines.

- Advantages: a) Lightweight; b) easy to install; c) economical; d) Good corrosion resistance; e) Smooth surface reduces friction losses; f) Long pipe sections reduce leakage/ infiltration potential; g) Flexible

- Disadvantages: a) Susceptible to chemical attack especially solvents; b) Strength affected by sunlight; c) Not suitable for above ground installations



# MDPE (Medium Density Polyethylene) Pipes

- being used for consumer connection pipes as an alternative to GI pipes
- MDPE pipes are available in sizes ranging from 20 mm to 110 mm dia for pressure class of PN3.2, PN4, PN6, PN10 and PN16



- Applications: a) Water distribution for town, rural & irrigation projects; b) Cold water plumbing reticulation; c) Household water connections from the main supply; d) Compressed air lines
- Advantages: a) Very smooth inner surface ensures no scaling and choking, b) Has less friction loss and gives better flow at lower heads; c) Easy to transport and store as the pipes available in 100, 200 and 300 metre coils; d) No wastage of pipe as it can be cut to requirement at site; e) Less number of joint as the pipe is flexible and easy to repair, f) Easy tapping with specialty tapping joints, g) Tools-off installation possible with precision made fittings

  Disadvantages: a) Susceptible to chemical attack especially solvents; b) Strength may be affected by sunlight

# GRP (Glass Fibre Reinforced Plastic) Pipes

- Five pressure classes of pipes namely, PN 3, PN 6, PN 9, PN 12 and PN 15 correspond to the working pressure ratings of 3.06, 6.12, 9.18, 12.24 and 15.30 kg/sq.m, respectively
- Stiffness is the prime design criteria in the case of underground pipes
- GRP pipe stiffness is classified into four classes, i.e., A, B, C, D, depending on the type of installation, overburden above the crown of the pipe and the soil conditions
- GRP pipes are widely used in other countries where corrosion resistant pipes are required at reasonable costs

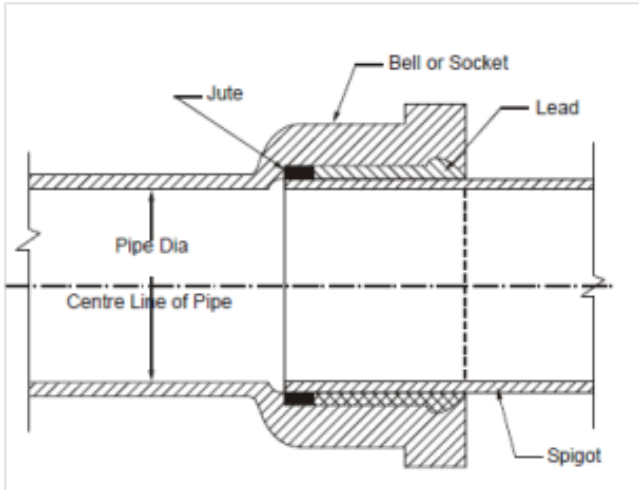
- Advantages: a) High strength to weight ratio; Corrosion resistant; b) Light weight compare to metallic and concrete pipes; c) Longer length and hence minimum joints enable faster installation
- Disadvantages: a) High material cost; b) Brittle, require careful installation; c) High installation cost



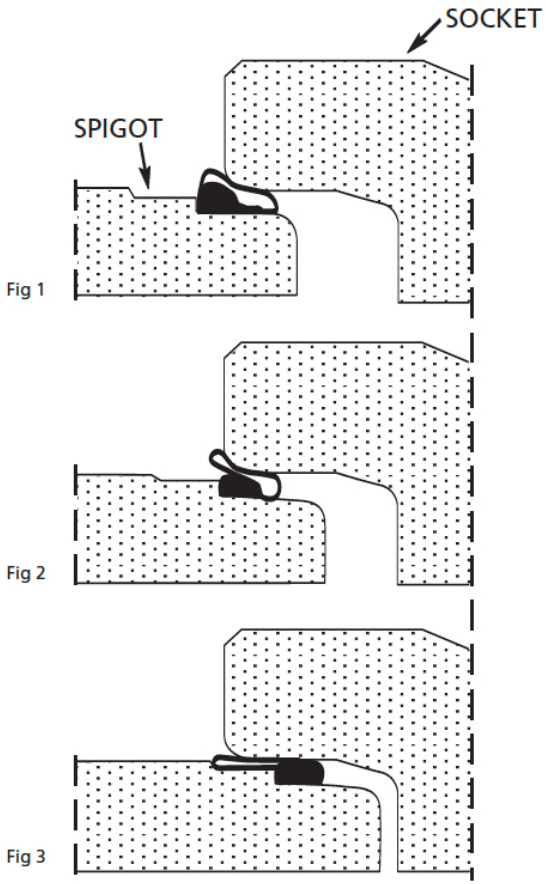
Pipe Joints

Pipelines are often long and pipe networks are also huge. Pipes are available in finite lengths. They need to be jointed to develop long pipelines and networks. The basic requirement of pipe joints are: a) Water tightness b) Strong enough c) Durability d)Economy e)Workability. There are various types of pipe joints

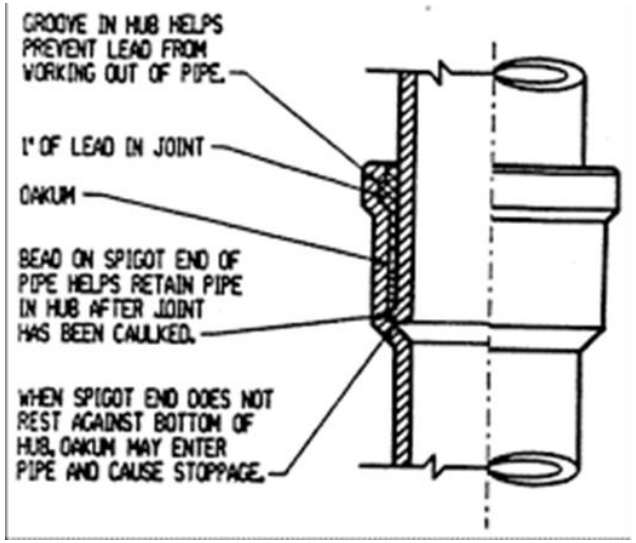
Bell and Spigot (Socket and Spigot Joints)

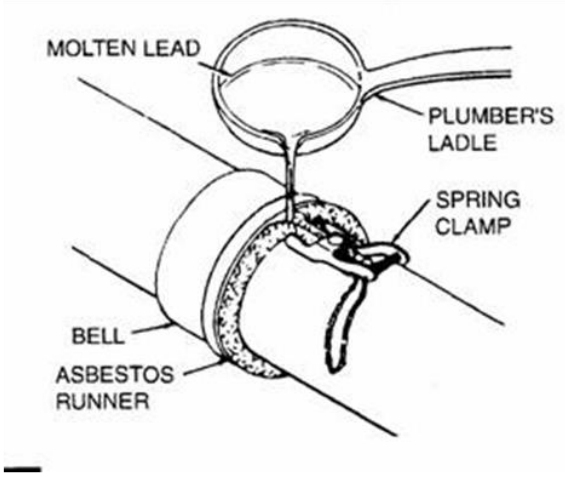




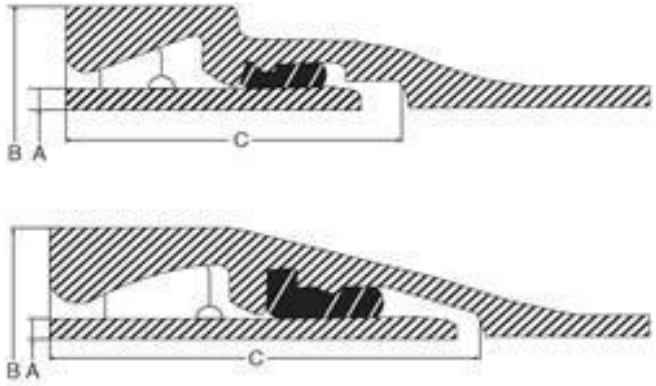
## Lead-caulked joints (bell and spigot)




This was very popular in earlier days. However, they are discontinued because of concerns of toxicity caused by lead in the drinking water.

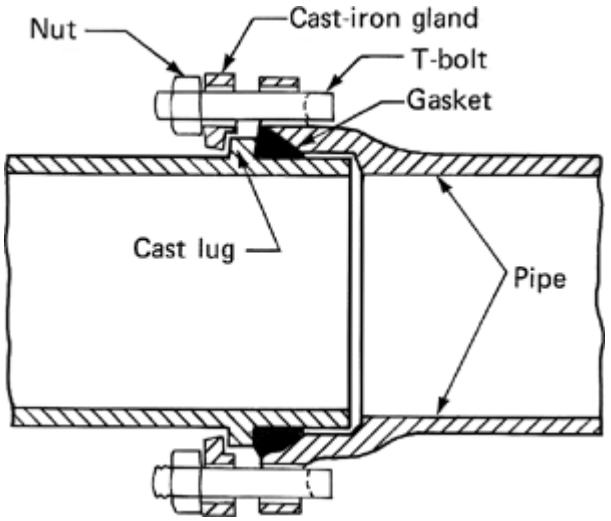
## Push-on flexible joints (bell and spigot)





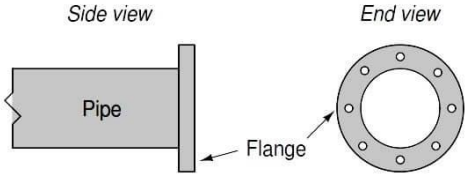

## Mechanical joints (bell and spigot)

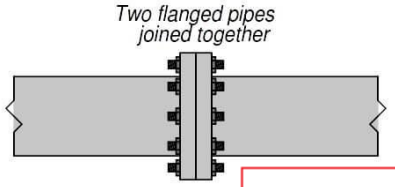
This type of joint is used when cast iron pipes are joined. A special type of metallic collar is fitted in the spigot portion around the cast lug and is tightened against the bell portion using nut and bolt. In order to make it water tight gaskets are often used.



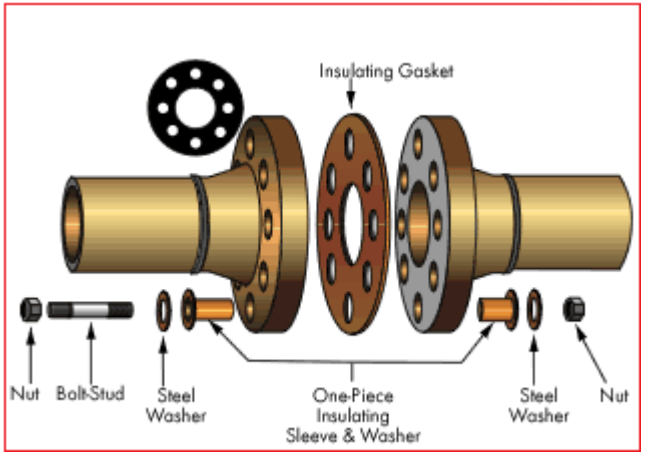

## Flanged Joints

Most common type of pipe joint used in the steel piping system. For cast iron pipe system, the pipes come with flanges at the both ends. For mild steel or carbon steel pipes, the flanges are either screwed down or welded at the ends. Often a rubber or leather gasket is used to make the joints water tight. This joints are rigid, and hence cannot be used where deflections or vibrations may be expected.



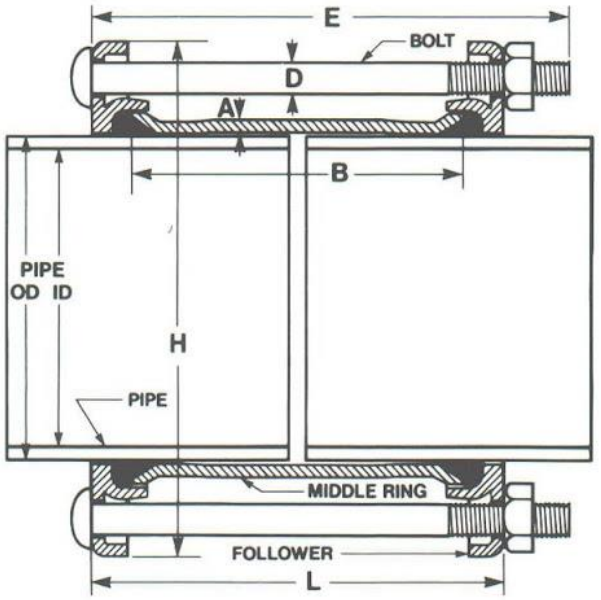





## Dresser Coupling

Dresser coupling is also a type of mechanical joint. It is used when two pipes with plain end need to be joined. A special type of mechanical collar is fitted and tightened over the ends. A gasket and an iron ring are slipped over each of the abutting ends of the pipes and a iron sleeve is inserted between the gaskets as a pipe stop. The iron rings are tightened by means of bolts. These joints are strong and rigid, and can withstand vibrations.




## Victaulic Coupling

Victaulic coupling is a special type of coupling to join pipes of materials and across multiple fire protection, industrial and commercial applications. These joints are proprietary and can be made rigid, flexible, etc. so suit the application.

